

EmbracePlus

USER MANUAL

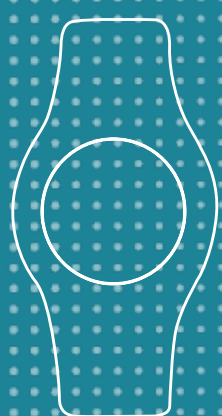
Table of contents



01 Embraceplus At A Glance	3
02 Using Embraceplus For The First Time	7
03 Using Embraceplus In Daily Life	15
04 Troubleshooting	27
05 FAQs	33
06 Regulatory And Compliance	38
07 Document History	73

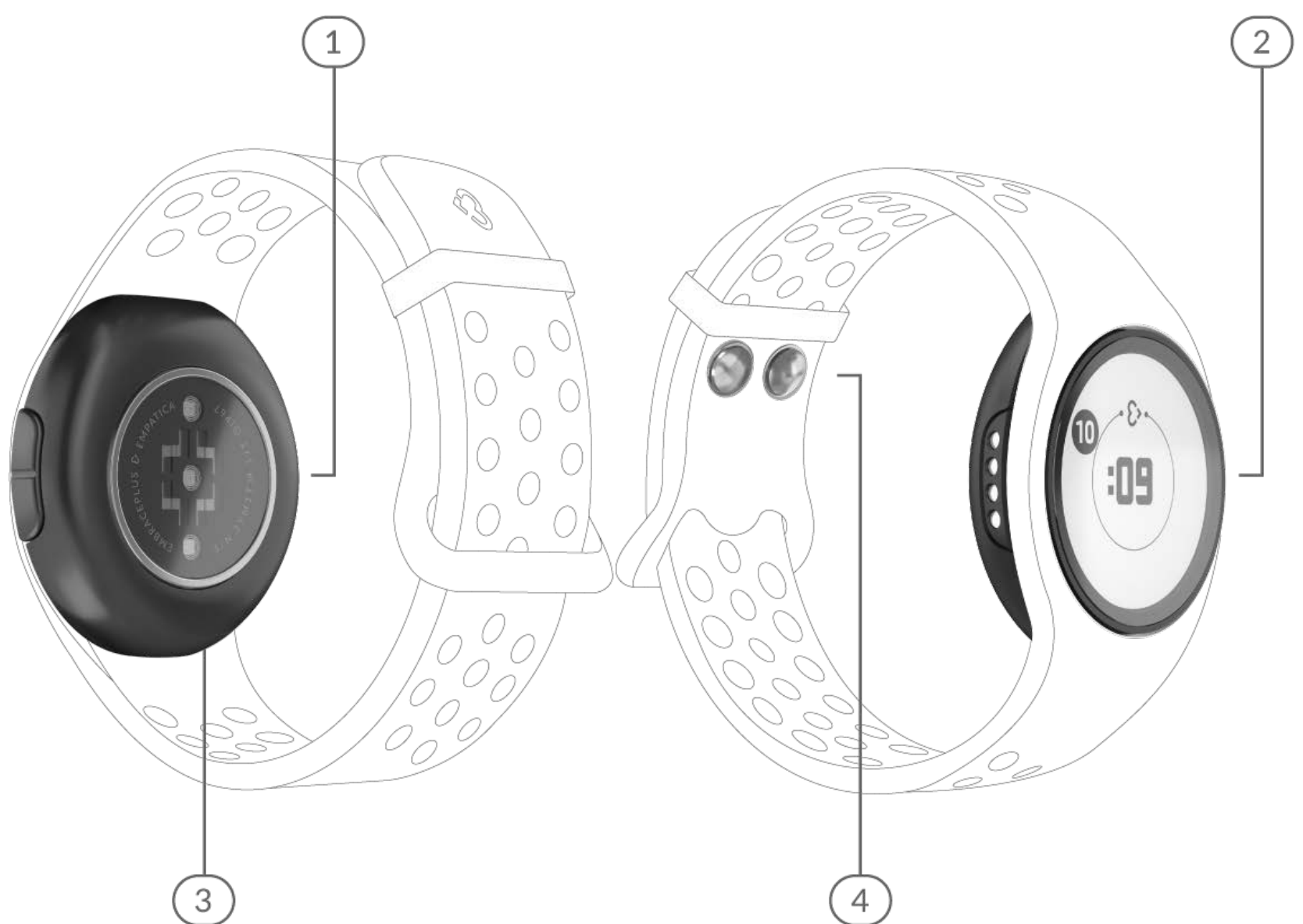
01

EmbracePlus
at a glance



What the EmbracePlus does

EmbracePlus is a wearable device that collects, processes, stores, and wirelessly transmits physiological parameters to a companion app/system. EmbracePlus runs different on-board algorithms to continuously analyze acquired data and extract specific digital biomarkers.



1. Serial number
2. E Ink display
3. Pod Case sensors: Accelerometer, Gyroscope, Photoplethysmogram (PPG), Temperature
4. Band sensors: Electrodermal Activity (EDA)

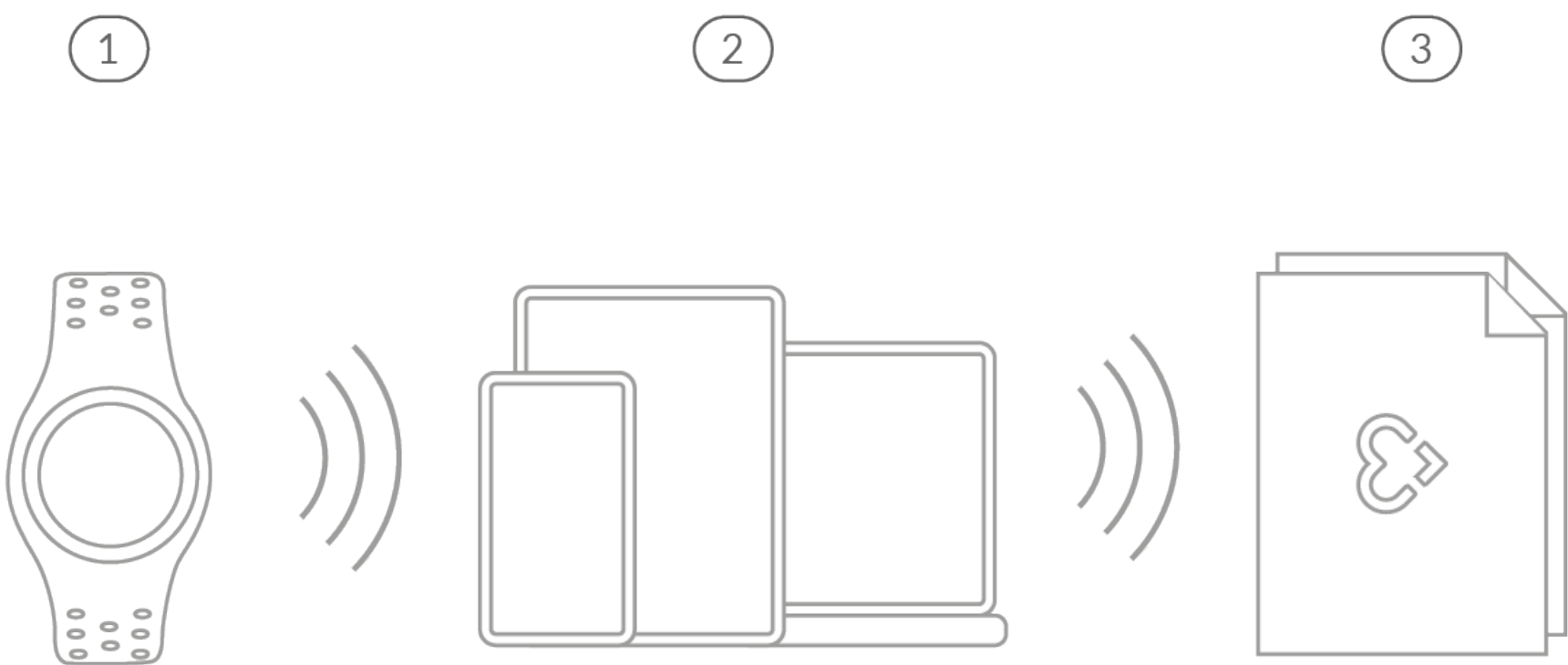
How it works

This is how EmbracePlus works:

The EmbracePlus (1), the wearable, collects physiological parameters from your wrist.

The data are then transferred to a companion app/system (2) via Bluetooth.

Data access (3) is then provided via the companion app/system for the intended use.



Types of data that EmbracePlus collects

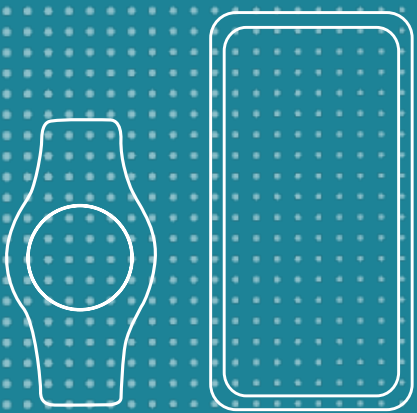
EmbracePlus seamlessly collects the following* physiological parameters from your wrist: pulse rate, pulse rate variability, respiratory rate, blood oxygen saturation, electrodermal activity, skin temperature, activity counts, and sleep.

*Depending on sensor configuration



02

Using EmbracePlus
for the first time



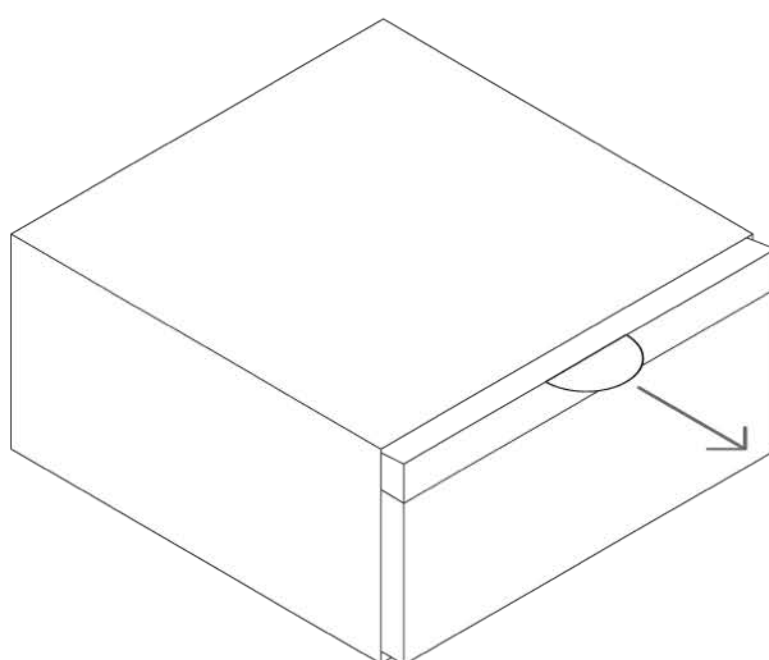
Getting started

Using EmbracePlus is simple. Follow these steps to get started:

STEP 1: UNBOX EMBRACEPLUS

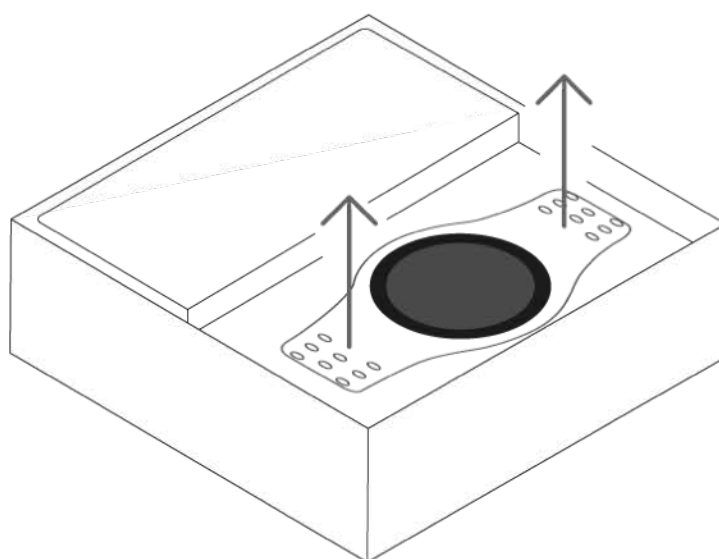
- a. Pull the tab and slide the sleeve from the package to extract the box.

a



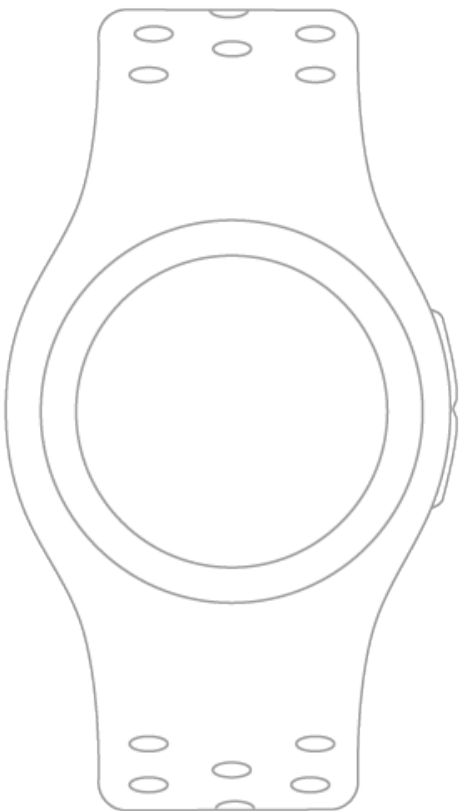
- b. Gently pull out the EmbracePlus.

b

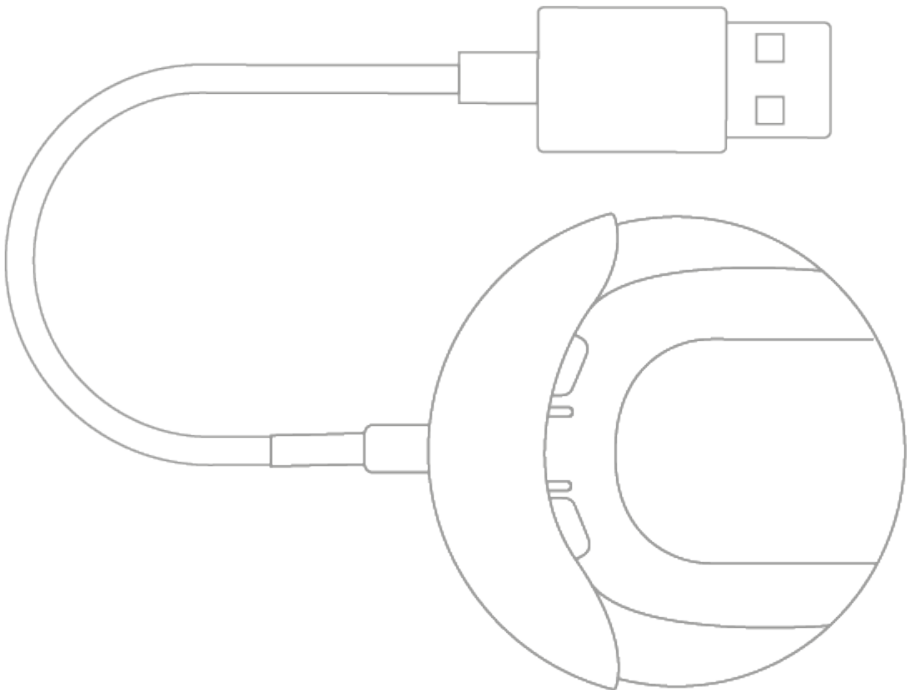


c. Components of the box:

I. EmbracePlus



II. Cable charger

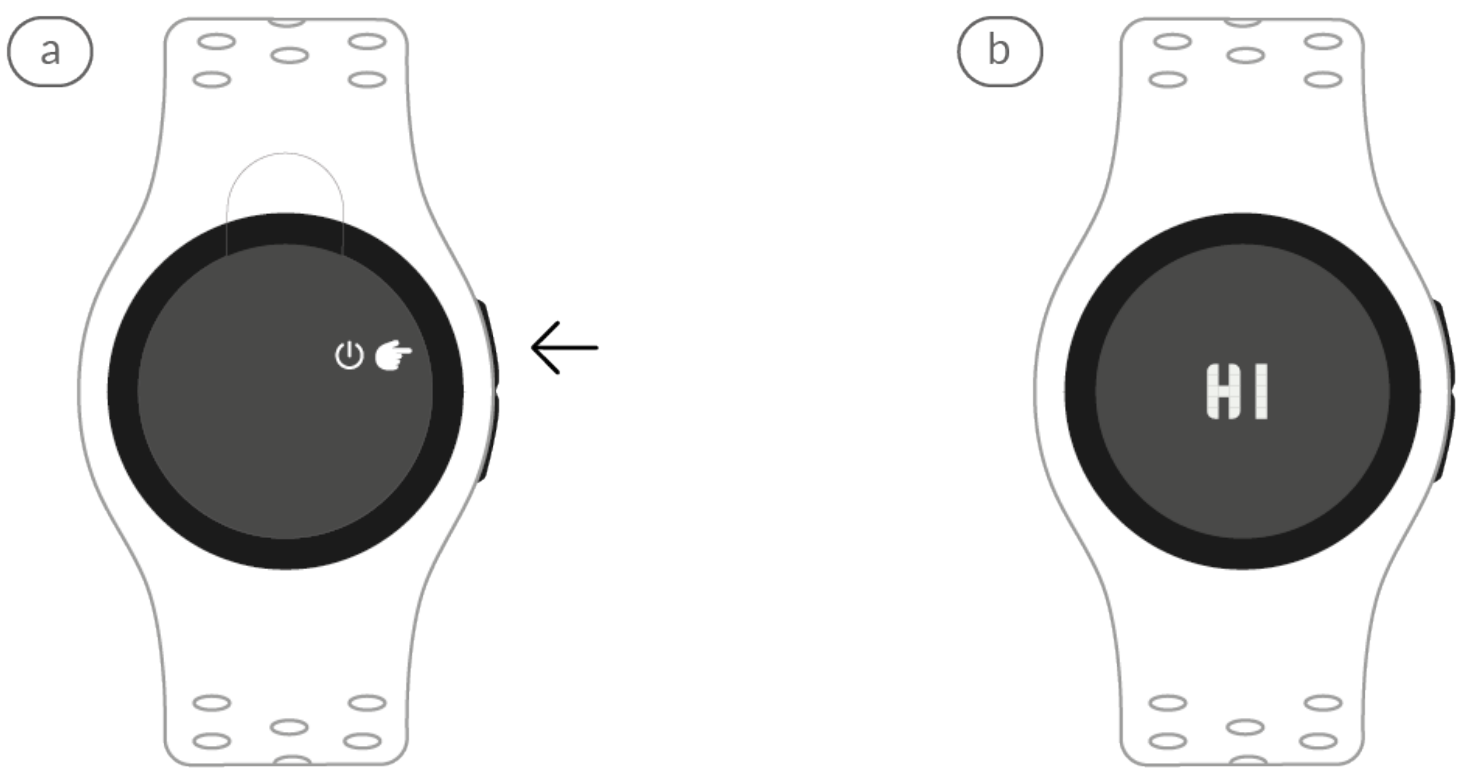


III. Folder with the Quick Reference Guide

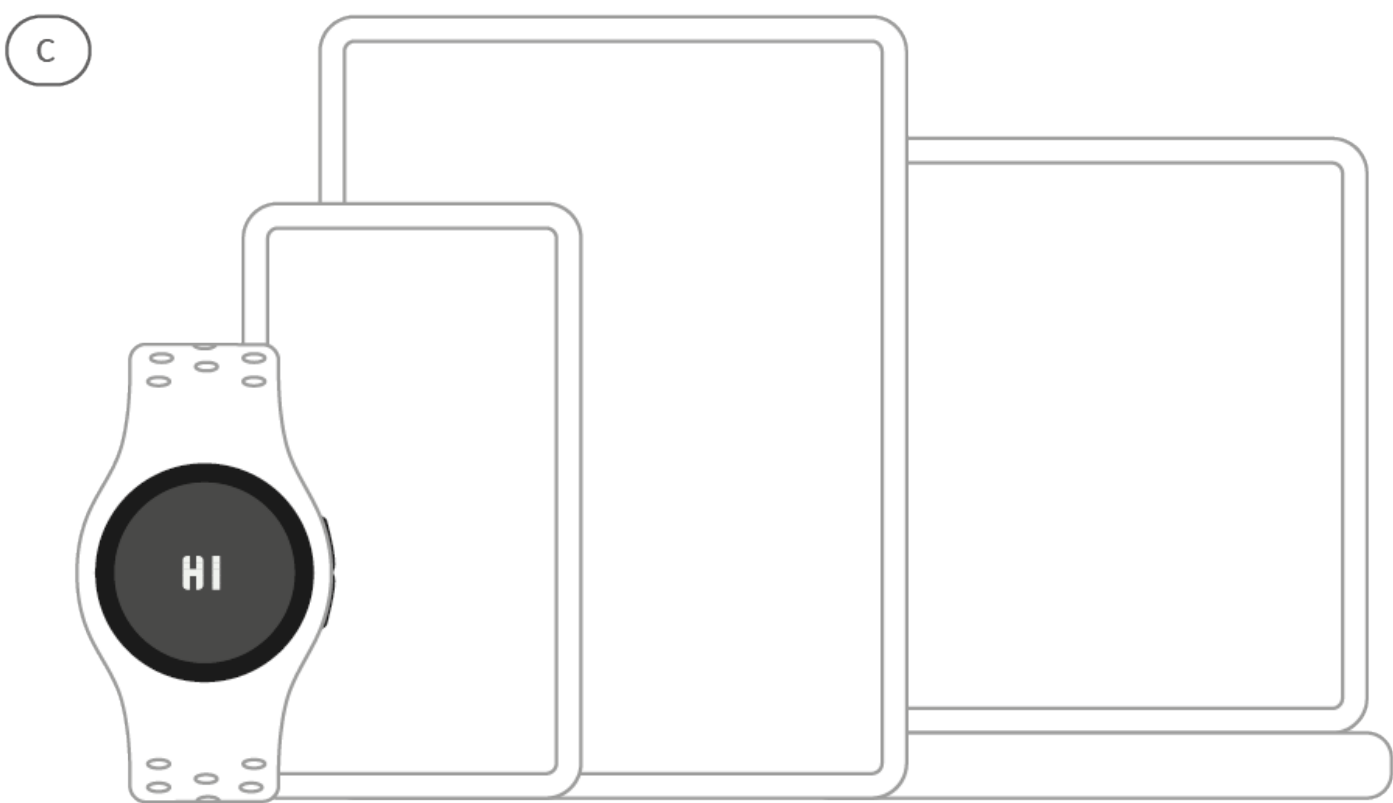


STEP 2: TURN ON EMBRACEPLUS

- a. Turn on EmbracePlus by pressing the upper button, as indicated by the small hand on the sticker.
- b. EmbracePlus will greet you with a HI. If nothing happens, you may need to charge it. Remove the sticker and proceed.



- c. Pair EmbracePlus to the companion app/system.



- If no pairing happens within a few minutes, EmbracePlus will automatically switch off to preserve battery. Return to step 1.*
- e. You are all set to use EmbracePlus. The administrator will give you further instructions on how the EmbracePlus should be used in your daily life.



IMPORTANT

If EmbracePlus doesn't turn on

In the unlikely event in which your EmbracePlus is out of battery, the device will not react when pressing the button. To turn it on, you will first need to charge EmbracePlus with the cable charger included in the box.

Please follow the charging instructions in the: ['How to Charge EmbracePlus' section](#).

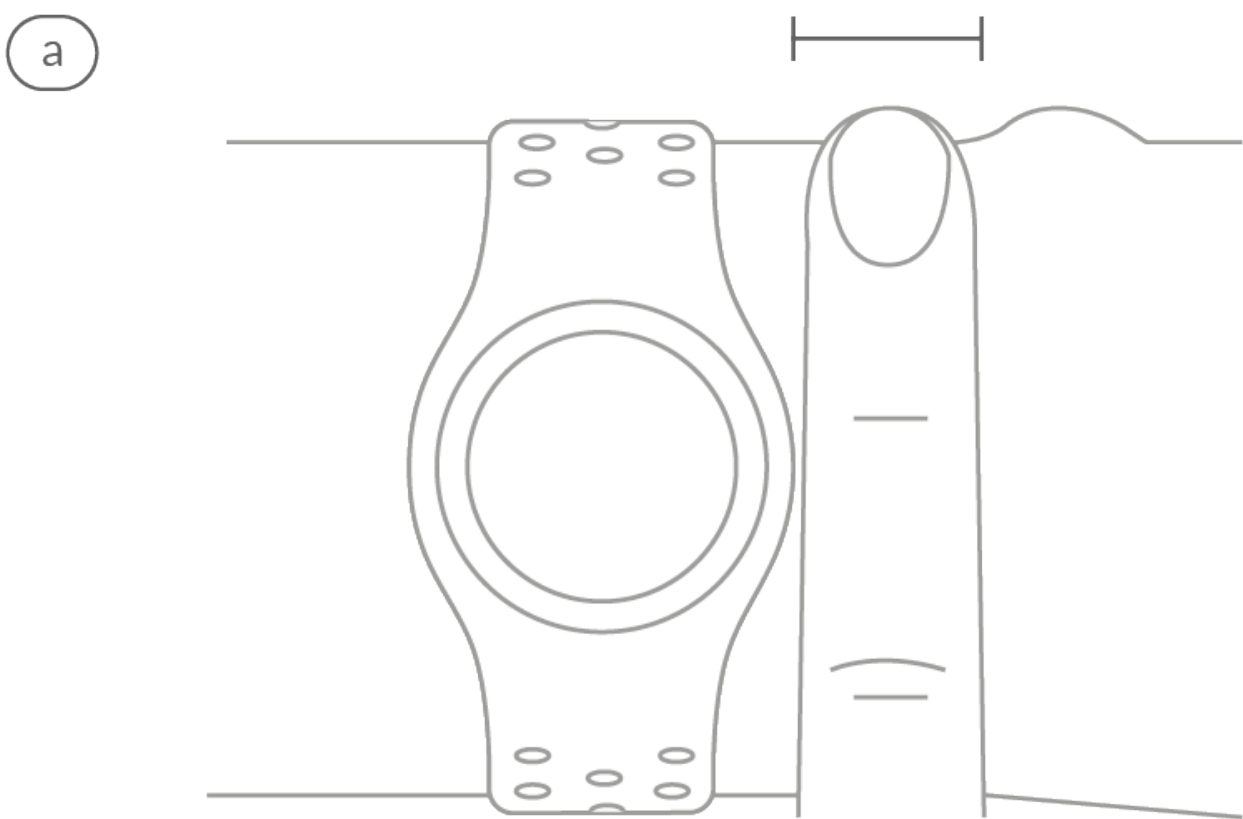
STEP 3: WEAR EMBRACEPLUS AND START RECORDING PHYSIOLOGICAL PARAMETERS

Wearing EmbracePlus correctly is crucial for enabling proper recording of physiological parameters. The EmbracePlus needs to be correctly in place with the sensors touching the skin in order for it to collect data properly.

The best way to wear it is by following the steps below:

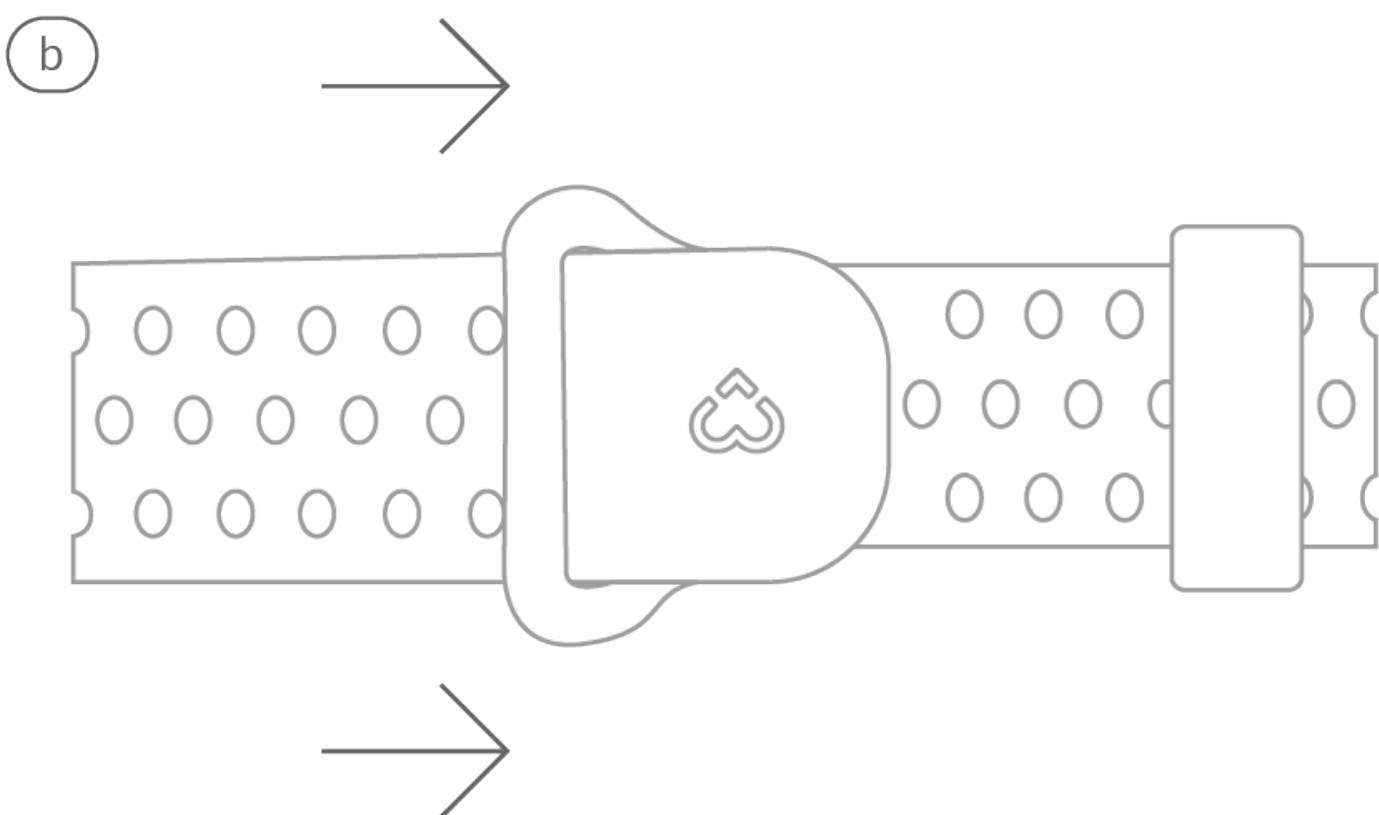
a. Place EmbracePlus on your wrist

Wear it on your non-dominant hand (unless otherwise specified by the administrator) for more accurate monitoring. Your device should be a finger’s width from your wrist bone.



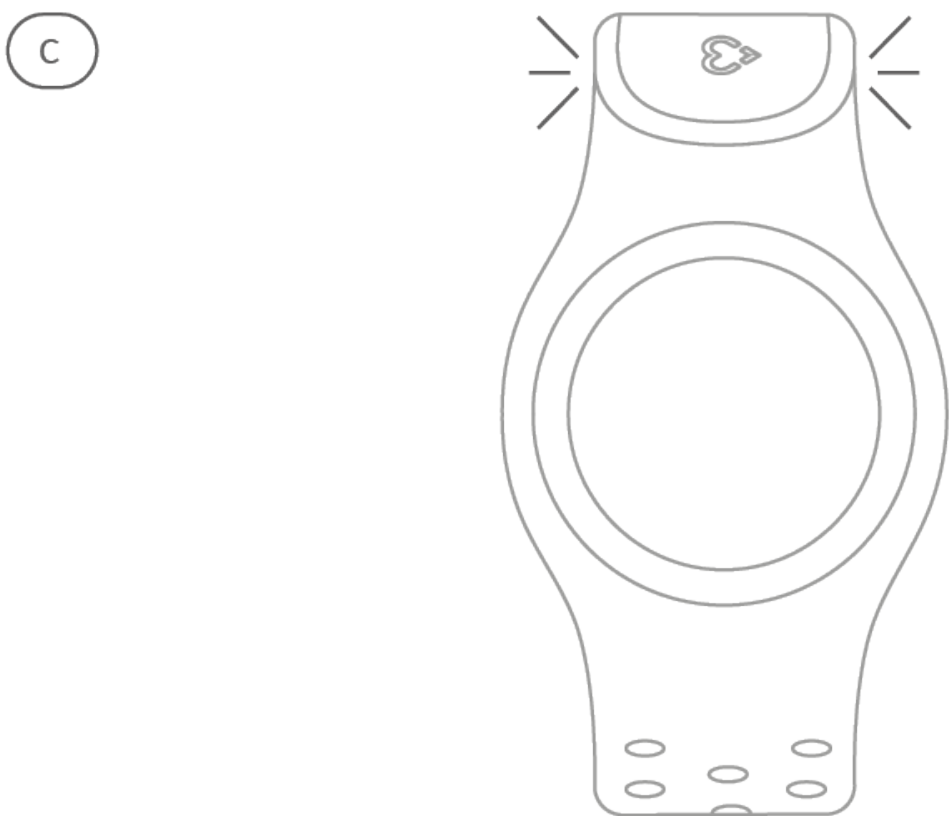
b. Adjust the band fit

Insert the band inside the loop and pull until you feel a soft pressure on your wrist. Make sure it doesn’t easily change position.



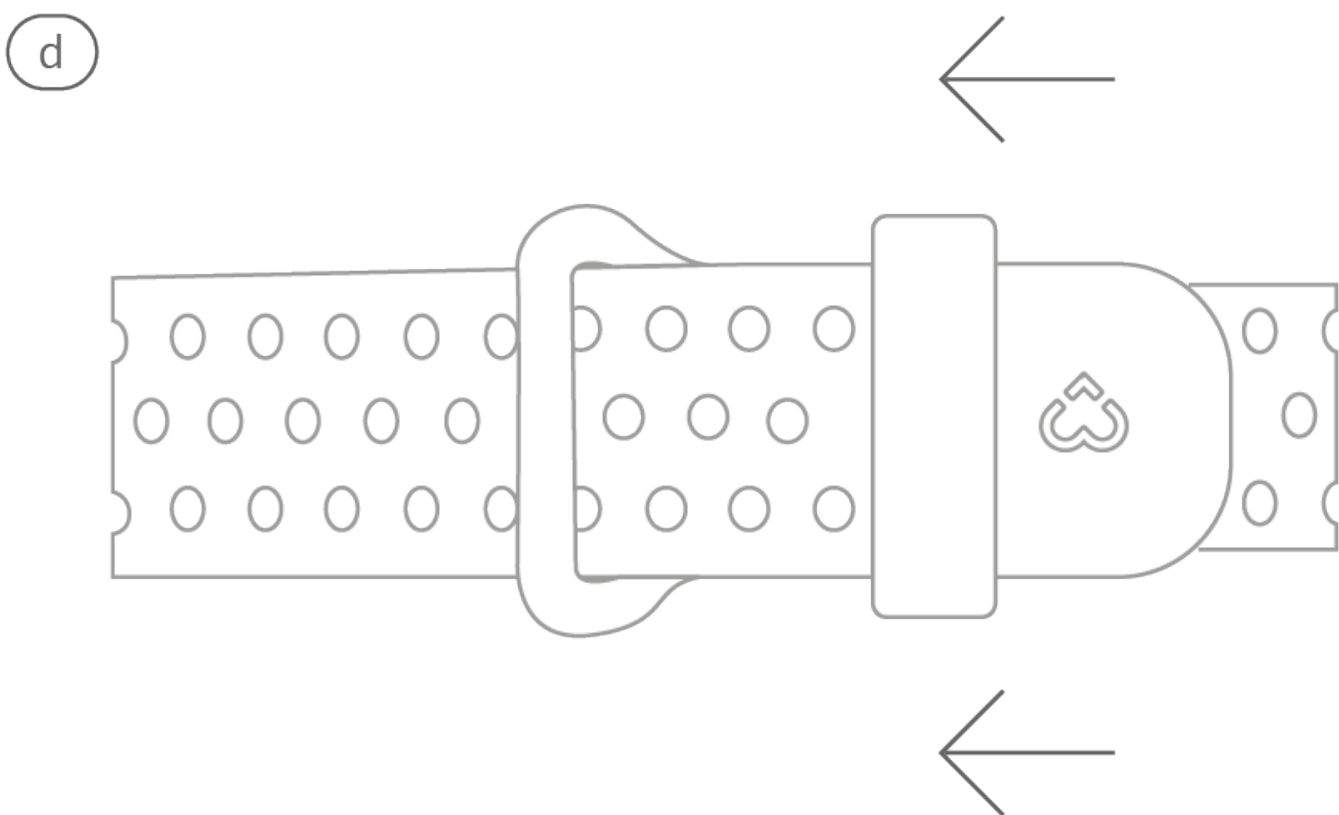
c. Fasten the band

Push the pins into the holes. Be aware that a loose band might impair data collection.



d. Secure using the loop

Slide the tail of the band through the loop to secure it in position.





IMPORTANT

Make sure the EmbracePlus fits snugly but not too tightly. Only wear the EmbracePlus on intact skin. Do not use the EmbracePlus with injured skin or other types of skin diseases located in the area where you plan to wear the EmbracePlus. Do not use the EmbracePlus if it appears to be damaged.

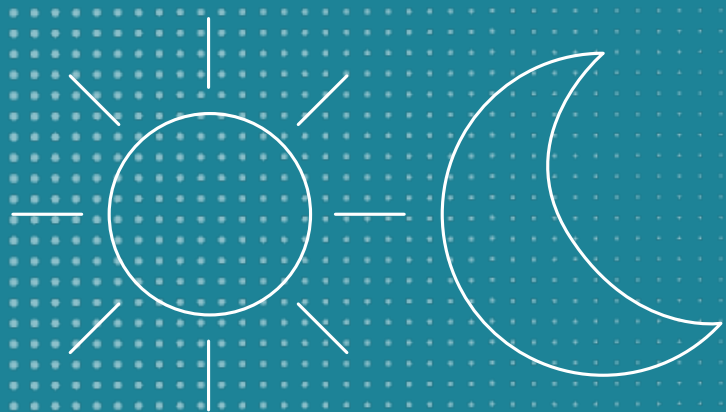
EmbracePlus is waterproof and can be worn while exercising or taking a shower. However, Embrace Plus should be removed while swimming in saltwater or while using a hot tub, sauna, or steam room.

In case of skin irritation after prolonged use, it is suggested to switch the recording location, by placing the EmbracePlus on your other wrist. Also make sure to [clean the EmbracePlus](#) to avoid further irritation. In case of an allergic reaction persisting beyond 2-3 days, it is suggested to immediately stop using the device and contact the administrator.

The materials used in the manufacturing of the EmbracePlus are widely used in wearable devices. Empatica performed biocompatibility tests according to the ISO 10993 standard series and the device successfully passed those tests.

03

Using EmbracePlus in daily life



Display and interaction guide

The EmbracePlus display uses E Ink technology, which consumes little energy and always remains on, even if the EmbracePlus is out of battery.

When EmbracePlus is recording data and connected to the companion app/system, you will find the following components on the display:

1. *Time: Hours*
2. *Time: Minutes*
3. *Bluetooth connection with companion app/system:
Empatica heart*



HOW TO READ THE TIME

EmbracePlus has a 12-hour clock, with no distinction between AM and PM. The hour is shown in the outer circle, while the minutes are shown in the center of the display. The time indicated by EmbracePlus is based on the time of the paired companion app/system.



HOW TO CHECK IF EMBRACEPLUS IS RECORDING DATA

When EmbracePlus is recording data, the display background is light.

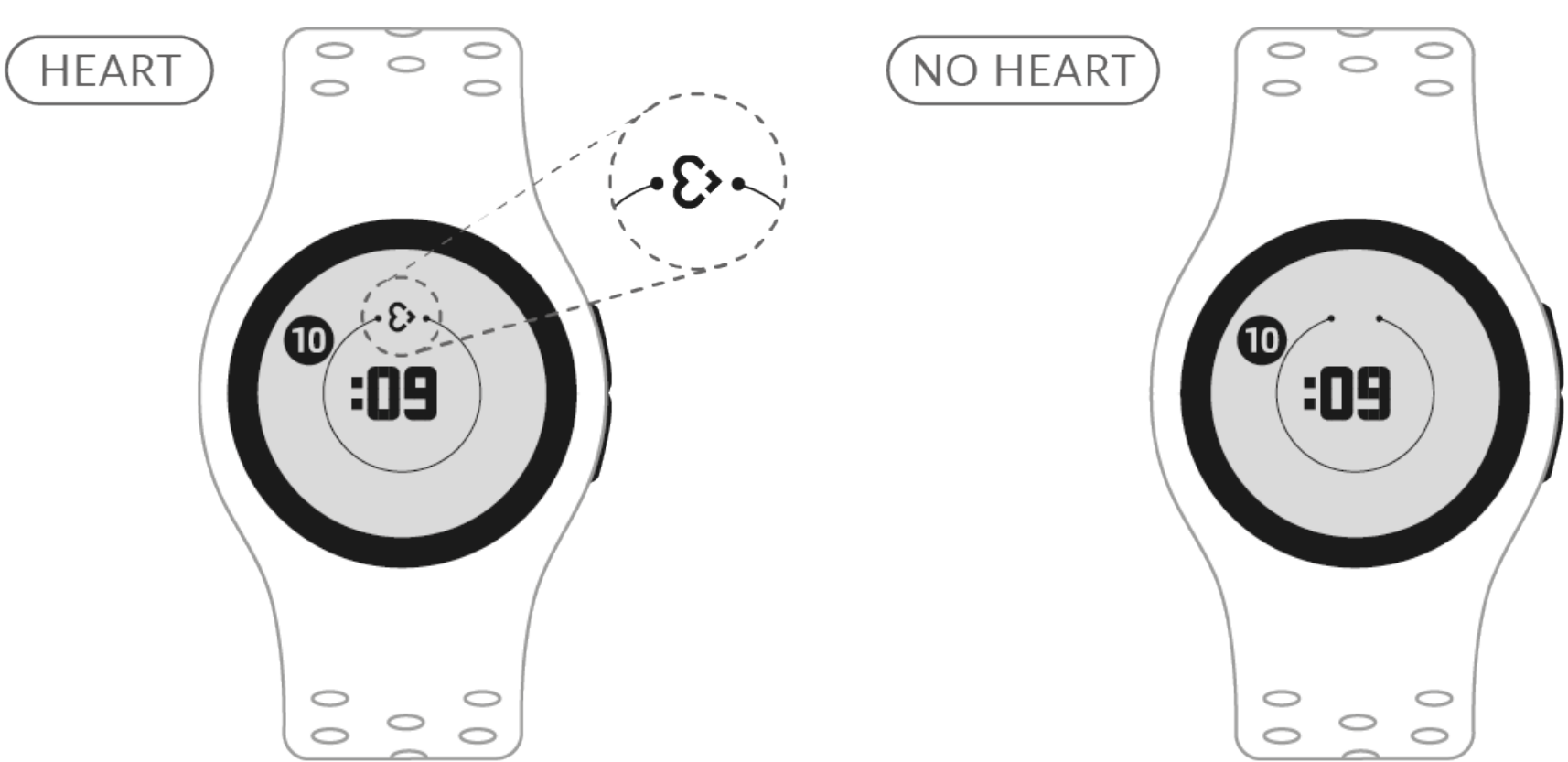


When EmbracePlus is not recording data, the display background will turn dark. This behavior is to be expected in the following cases:

- When EmbracePlus is charging
- For a few seconds after it is removed from the cable charger to calibrate
- For a few seconds when starting up or rebooting
- [Specific error cases](#)

HOW TO CHECK IF EMBRACEPLUS IS CONNECTED TO THE COMPANION APP/SYSTEM

The Empatica heart will be visible if EmbracePlus is connected to the companion app/system via Bluetooth, and not visible if it is disconnected. Although EmbracePlus can work stand-alone for several hours, the more it stays connected to the companion app/system, the better. This ensures that the recorded data are always synchronized.

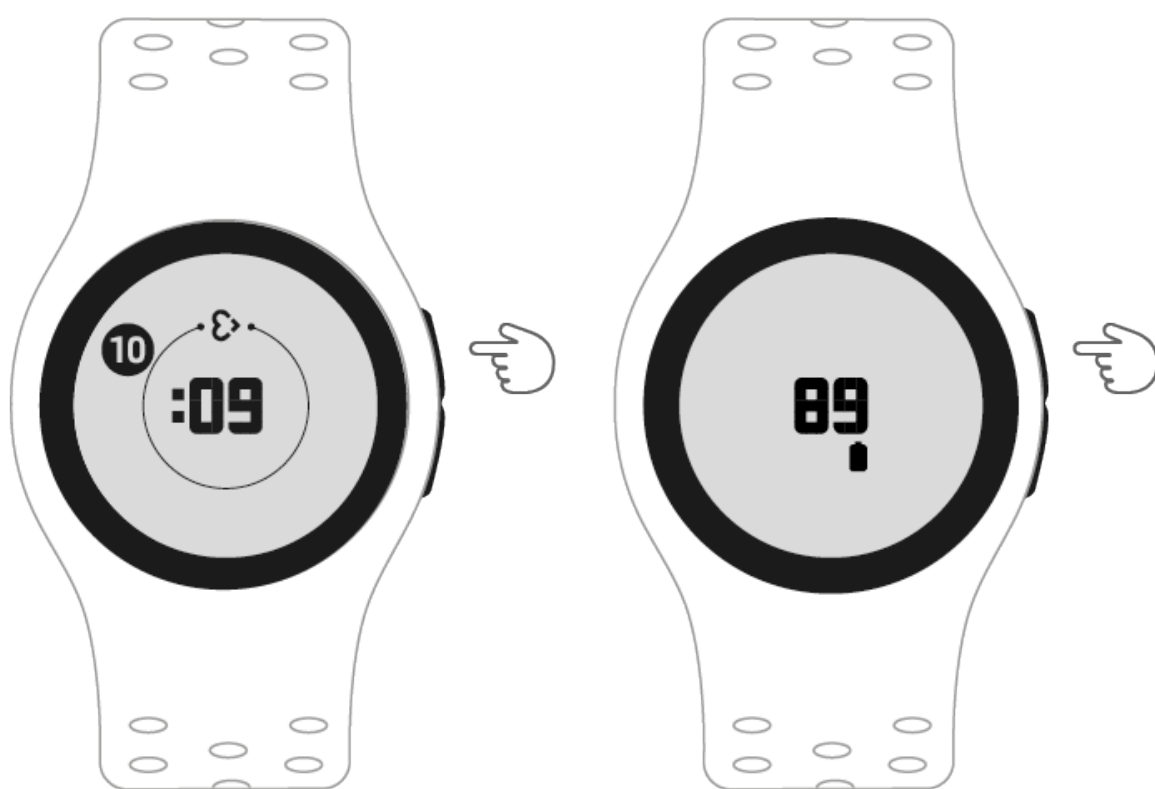


How to check and charge the battery

In order for it to continuously collect your data, EmbracePlus needs to be kept charged according to the instructions of the administrator. You can always check the status of your battery directly on EmbracePlus.

HOW TO CHECK THE BATTERY PERCENTAGE

By pressing one of the buttons on the right side of the EmbracePlus, the time will disappear, and the battery percentage will be shown. You can go back to the time by pressing one of the buttons, otherwise, the display will automatically switch back to the current time after 10 seconds.



LOW BATTERY - CHARGE SOON

When the EmbracePlus battery level begins to run low, the battery icon will be visible on the display.



LOW BATTERY - CHARGE NOW

When the EmbracePlus battery level is critical, a “low” message will appear on the display. The message can be dismissed by placing the EmbracePlus on the charger or pressing one of the side buttons.



OUT OF BATTERY - PLACE IN CABLE CHARGER

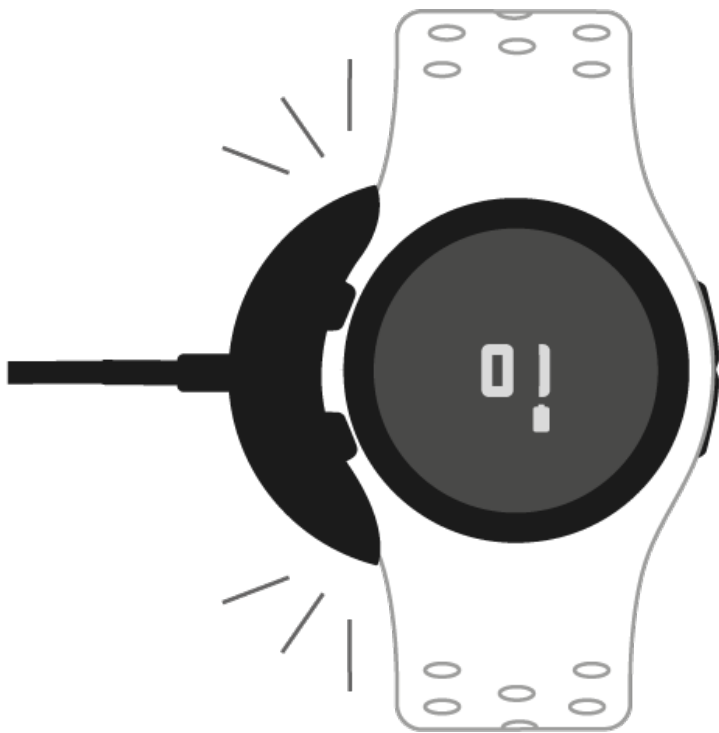
When completely out of battery, the EmbracePlus will turn off. The background of the display will turn dark to highlight that data recording has stopped and an empty battery icon will be displayed.



HOW TO CHARGE EMBRACEPLUS

When you need to charge EmbracePlus, follow these simple steps:

1. Use the cable charger included in the EmbracePlus package and plug it into a standard 5V USB 2.0 power source.



2. The cable charger and EmbracePlus will automatically attach through magnets. Make sure the display and side buttons are visible while you slide EmbracePlus into the cable charger. You will feel a slight pull and EmbracePlus will snap into place.
3. Once the charging starts, the display background will turn dark and show the available battery percentage. A full charge takes about 1 hour and 30 minutes.



4. When EmbracePlus is fully charged, gently remove it by holding the cable charger in place and sliding out the EmbracePlus.
5. For about 10 seconds, the EmbracePlus display will have a dark background. Once the display background turns light and the time is shown, EmbracePlus is up and running and ready to be used.

HOW TO TAG AN EVENT

The EmbracePlus allows the user to tag events according to the administrator’s instructions. To tag an event, press one of the side buttons for one second and a checkmark will appear on the display confirming the action. The display will then go back to show the time.



HOW TO CLEAN AND SANITIZE YOUR EMBRACEPLUS

It's recommended to clean your EmbracePlus every other week, or in case you experience itching or skin irritation. In order to clean the EmbracePlus, follow these six simple steps:

1. Remove EmbracePlus from your wrist.
2. Take a polyester/cellulose wipe.
3. Soak the wipe in 70% isopropyl alcohol for 5 minutes.
4. Thoroughly clean the EmbracePlus for at least 1 minute by wiping the whole device.
5. Visually check the whole device for traces of dirt. In case you see visible contamination, repeat the cleaning procedure.
6. Let EmbracePlus dry before wearing or sanitizing it.

In order to sanitize the EmbracePlus, follow these four simple steps:

1. Take a polyester/cellulose wipe.
2. Soak the wipe in 70% isopropyl alcohol for 5 minutes.
3. Thoroughly wipe the EmbracePlus for at least 90 seconds.
4. Let EmbracePlus dry before using it again.



IMPORTANT

Do not use cleaning agents other than those described above without explicit written instructions from Empatica.

Tips and tricks

While wearing the EmbracePlus watch make sure that:

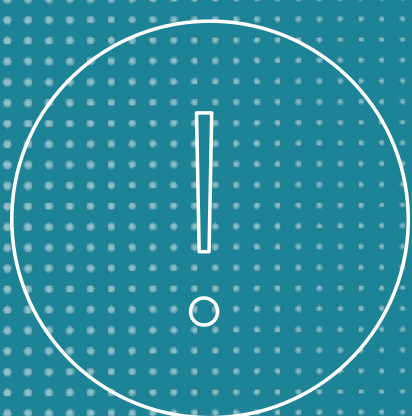
1. Bluetooth is ON on the companion app/system.
2. The companion app/system is within Bluetooth range of 10 meters/30 feet, free of any obstacles (i.e. walls, heavy fabric) to ensure timely data collection.

Wear the EmbracePlus preferably on your non-dominant wrist, or according to the instructions of the administrator.

It is important to always have EmbracePlus charged. When the EmbracePlus battery runs low, the battery icon will appear on the display.

04

Troubleshooting



NOT RECORDING - MEMORY FULL

When the display background is dark and an ‘M’ appears, the EmbracePlus memory is full of already recorded data, and it has stopped recording new data. A full memory occurs when the EmbracePlus is disconnected from the companion app/system for several hours and is unable to synchronize data.

Make sure Bluetooth is turned on and the companion app/system is within the Bluetooth range (10 meters / 30 feet) to connect the EmbracePlus. Once the Bluetooth connection is re-established, your EmbracePlus will restart data synchronization and record new data.



NOT RECORDING - OUT OF BATTERY

When EmbracePlus is out of battery, it cannot record data and therefore the display background will turn dark. Charge your EmbracePlus to start recording data again.



CALIBRATING

When the EmbracePlus is calibrating, you will see 4 dots moving on a dark background. This happens right after the EmbracePlus is removed from the charger. The watch face will be visible after a few seconds. Keep wearing your EmbracePlus.



NOT RECORDING - MISSING COMPANION APP/SYSTEM

When the background is dark and an ‘X’ appears, the EmbracePlus is missing the companion app/system connection. Make sure the companion app/system has Bluetooth activated and is within range (10 meters/30 feet) of the EmbracePlus to re-establish a connection.

Once the companion app/system connects, the display background will turn light and the Empatica heart will appear.



SUDDEN REBOOT

In case EmbracePlus self-reboots, you will see a dark display with ‘HI!’ for a few seconds. EmbracePlus self-reboots to restore proper functioning in cases of unexpected behavior.

During this brief interval, data recording will stop. Once recording resumes, the display background will turn light again and the time will be shown. Data synchronization will be automatic and no further action is required.



MANUAL REBOOT

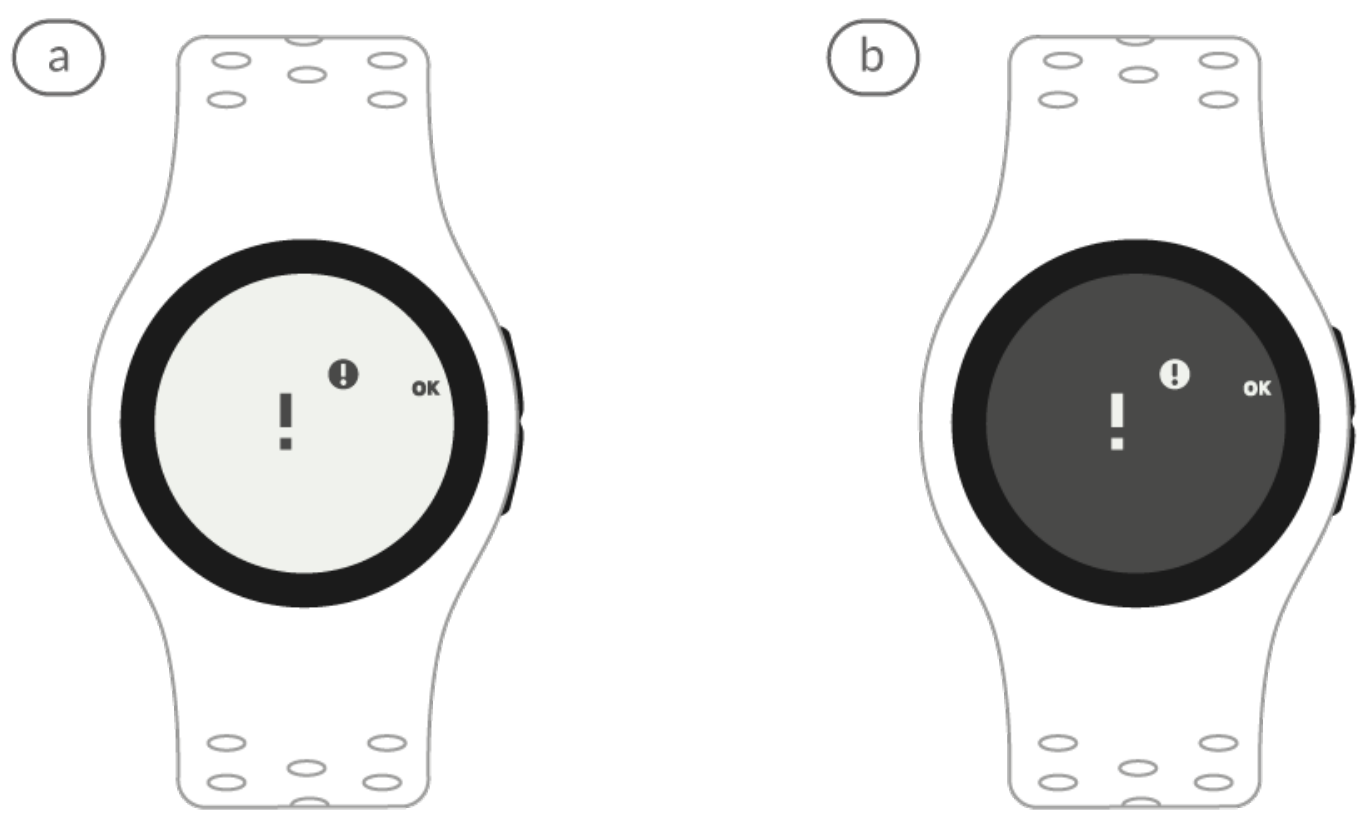
In case there is a firmware error detected by the EmbracePlus, an exclamation point (!) symbol will appear on the display, indicating that a manual reboot is required.

- a. The firmware error may happen while the EmbracePlus is recording data, shown by the light display background.
- b. The firmware error may also happen while the EmbracePlus is not recording data, shown by the dark display background.

Press the OK (upper) button and the EmbracePlus will reboot, confirmed by the ‘HI’ message. This should resolve the issue.

EmbracePlus will also automatically perform a reboot if in case the manual reboot is not confirmed within 10 minutes of pressing the OK button.

If the firmware error persists even after the reboot, please contact the administrator for further assistance.



NOT RECORDING - CRITICAL TEMPERATURE

When the EmbracePlus reaches an unusually high temperature (60°C), a ‘T’ symbol will appear on the display. To prevent battery and hardware damage, the EmbracePlus will automatically be put in hibernation mode.

Make sure that the environmental conditions are within the operational range (0 - 50°C) of the EmbracePlus. Then, press the OK (upper) button and the EmbracePlus will reboot and wake up from hibernation, indicated by the ‘HI’ message.

In case the critical temperature is detected again after waking up, EmbracePlus will re-enter the critical temperature state and automatically go into hibernation.

It is possible that, while in hibernation, the EmbracePlus will run out of battery. If the EmbracePlus doesn’t turn on after pressing the OK button, place it in the cable charger to charge.

Likewise, it is possible that the EmbracePlus reaches a temperature level below 0°C. When this happens, the display will freeze due to display limitations, showing the last visualization before the error occurred. However, EmbracePlus will continue to work as expected until -20°C.

When the temperature goes below -20°C, the EmbracePlus will automatically go into hibernation mode. Make sure that the environmental conditions are within the operational range (0 - 50°C) before pressing the OK (upper) button to reboot the EmbracePlus and wake it up from hibernation.



05

FAQs



1. HOW DOES THE SYSTEM WORK?

The watch collects physiological parameters from your wrist and sends it via Bluetooth to the companion app/system. Data are then made accessible for their intended use through the companion app/system.

2. DO I NEED TO INTERACT WITH EMBRACEPLUS?

As long as it is charged and worn snugly on your wrist, EmbracePlus will collect data automatically and require no interaction. To ensure smooth data collection, make sure you take it off to charge as indicated by the administrator.

Other than wearing the EmbracePlus correctly and keeping it charged, the administrator may also require that you tag events. Refer to the administrator's instructions for using the EmbracePlus in your daily life.

3. HOW CAN I QUICKLY CHECK IF THE EMBRACEPLUS IS ON AND TRANSMITTING DATA?

- Make sure the display background is light, which indicates that data are being recorded. If the background of the display is dark, data are not being recorded and you should open the companion app/system to diagnose the problem.
- Press one of the side buttons on the EmbracePlus to verify the battery levels.
- If the Empatica heart icon is visible, the EmbracePlus is synchronizing data via Bluetooth to the companion app/system.
- Additionally, you can check the bottom of the EmbracePlus (the part that is in contact with your wrist). When EmbracePlus is on, the light sources under the device are likewise on. If the lights are off, then the EmbracePlus is likely to be not recording data as well.

4. HOW CAN I TURN THE EMBRACEPLUS ON/OFF?

Once you unbox EmbracePlus and turn it on, it will remain ON until it runs out of battery. In case it runs out of battery (you will see a battery icon on a dark background), please place it back in the cable charger and it will automatically switch ON.

5. HOW CAN I READ THE TIME ON EMBRACEPLUS?

The outer ring of the display shows the hours (12h) and the central section shows the minutes.

6. WHAT DATA DOES THE WATCH RECORD FROM MY WRIST?

The EmbracePlus records the following physiological parameters*:

- Pulse Rate (PR)
- Pulse Rate Variability (PRV)
- Respiratory Rate (RR)
- Blood oxygen saturation (SpO2)
- Temperature (TEMP)
- Electrodermal Activity (EDA)
- Activity Counts
- Sleep

*Depending on sensor configuration

7. IS THE EMBRACEPLUS WATERPROOF?

EmbracePlus is waterproof. You can wear it while exercising or taking a shower, but please remove it while swimming in saltwater or while using a hot tub, sauna, or steam room.

8. WHY HAS THE DISPLAY OF MY EMBRACEPLUS TURNED DARK?

The display background of the EmbracePlus turns dark when data recording stops. This happens when you take off your EmbracePlus and put it on the cable charger, or if the EmbracePlus battery has drained.

9. HOW CAN I CHECK THE BATTERY LEVEL OF THE EMBRACEPLUS?

The battery level of the EmbracePlus can be verified by **pressing one of the buttons** on the right-hand side of the EmbracePlus. The battery level will then appear on the display. The display will go back to showing the time when a button is pressed again. If no button is pressed, the display will go back to showing the time after 10 seconds.

10. HOW WILL I KNOW WHEN TO CHARGE?

EmbracePlus will show a battery icon on the bottom right of the display when the battery is low. This icon will stay there until the EmbracePlus is placed in the cable charger.

When the battery level is critical, a message saying “low” will appear on the center of the display. You should place the EmbracePlus in the cable charger as soon as possible.

11. I MIGHT BE ALLERGIC TO A SPECIFIC MATERIAL. WHAT IS EMBRACEPLUS MADE OF?

- POD Case: Polycarbonate
- Thermal Ring: Stainless Steel 316L
- PPG window: Gorilla Glass 3
- POD Buttons: Thermoplastic Polyurethane
- E Ink Display: Gorilla Glass
- Band: Biocompatible silicone rubber
- EDA electrodes: Stainless Steel 316L
- Buckle: Stainless Steel 316L

12. WHAT SHOULD I DO IF MY SKIN IS RED, ITCHY, OR I FEEL PAIN?

Stop wearing EmbracePlus if the skin on your wrist becomes red, itchy, or if any pain is felt. If this happens, please contact the administrator with photos of the affected area.



IMPORTANT

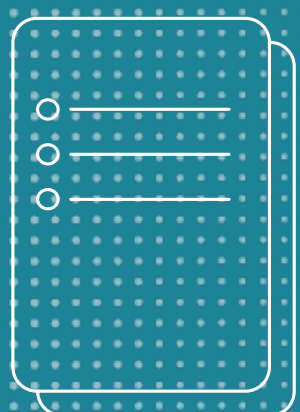
EmbracePlus should only be worn on the surface of healthy skin. Without regular cleaning, skin irritation is more likely to occur, so please make sure to [clean your EmbracePlus](#) regularly.

13. HOW CAN I RECEIVE MORE ASSISTANCE?

Contact the administrator. Unless specifically instructed to, please do not contact Empatica directly.

06

Regulatory and Compliance



EmbracePlus product labeling

The information directly printed on the device itself is presented in Figures 1 and 2.



Figure 1 - Back view of EmbracePlus

On the back of the device, the following information is displayed:

- Product Name and Manufacturer name: EmbracePlus by EMPATICA
- Device Serial Number: S/N 123456789
- IP Classification: Water Resistant (IP67)

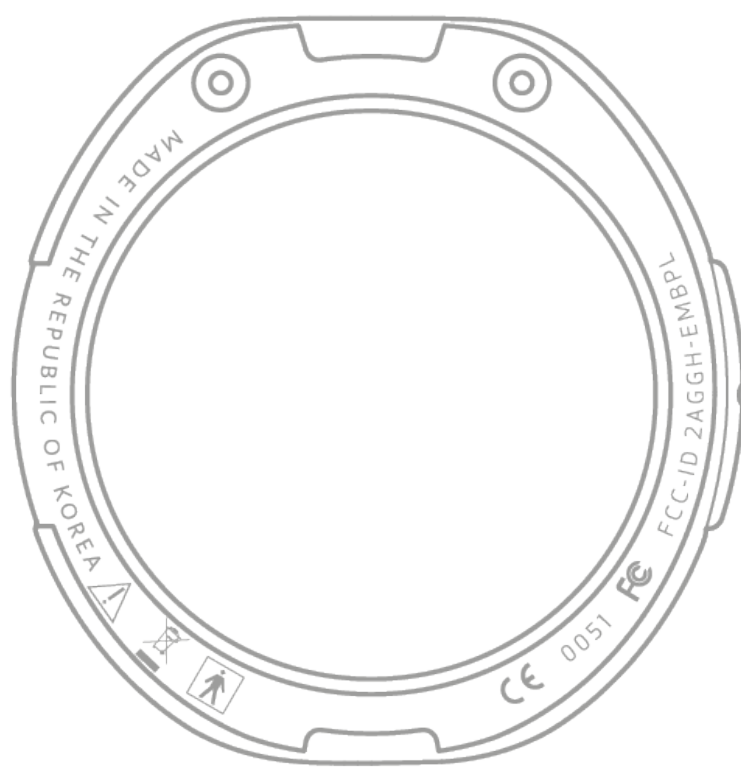


Figure 2 - Drawing representing the front view of EmbracePlus with band detached

On the front of the device the following information is displayed, note that the information is visible after detaching the band:

- Place of manufacturing: Made in Korea
- FCC symbol and FCC ID: FCC-ID 2AGGH-EMBPLUS
- CE Mark and identification of Notified Body: CE 0051
- Applied Parts type BF symbol
- Precaution Symbol
- WEEE Symbol

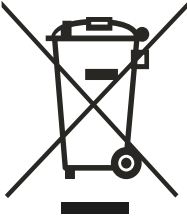
EmbracePlus laboratory testing

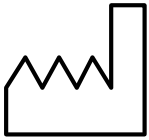
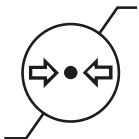
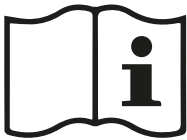
EmbracePlus has been subject to specific laboratory testing to assess its safety, electromagnetic compatibility, usability and biocompatibility. Tests have been performed according to the following standards:

- EN 60601-1:2006/A1:2013 (IEC 60601-1:2005/A1:2012): Medical electrical equipment – Part 1: General requirements for basic safety and essential performance
- EN 60601-1-2:2015 (IEC 60601-1-2:2014): Medical electrical equipment - Part 1-2: General requirements for basic safety and essential performance - Collateral Standard: Electromagnetic disturbances - Requirements and tests
- IEC 60601-1-6:2010 + Amd1:2013: Medical electrical equipment - Part 1-6: General requirements for basic safety and essential performance - Collateral standard: Usability
- IEC 60601-1-11:2015 Medical electrical equipment - Part 1-11: General requirements for basic safety and essential performance - Collateral standard: Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment
- IEC 62366-1:2015 Medical devices — Part 1: Application of usability engineering to medical devices
- ISO 10993-1:2018 Biological Evaluation of Medical Devices - Part 1: Evaluation and Testing within a risk management process
- EN ISO 10993-5:2009 Biological evaluation of medical devices — Part 5: Tests for in vitro cytotoxicity

- ISO 10993-10: 2010 Biological evaluation of medical devices – Part 10: Tests for irritation and skin sensitization
- FCC 47 CFR Part 15 Radio Frequency Devices
- IEC 62133-2: 2017 Secondary cells and batteries containing alkaline or other non-acid electrolytes - Safety requirements for portable sealed secondary lithium cells, and for batteries made from them, for use in portable applications - Part 2: Lithium systems

SYMBOLS USED IN LABELING

	Indicates compliance with the European Medical Device Directive 93/42/EEC. 0051 is the identification number of the Notified Body responsible for the CE Certification Process
	Indicates Empatica S.r.l as the legal manufacturer of EmbracePlus
	Indicates that the electromagnetic interference from the EmbracePlus is under the limits that are approved by the Federal Communications Commission
	The EmbracePlus uses Bluetooth Low Energy to transfer data.
	EmbracePlus is an electrical and electronic equipment and needs separate collection for disposal in accordance with Waste Electrical and Electronic Equipment Directive
	Indicates the device Serial Number
	EmbracePlus has two metal contact points that are used by the EDA sensor. This is type BF part which means that it may generate a leakage current when in contact with the skin. ISO 60601-1 safety test demonstrated that this part is safe.
	Indicates the need for the user to consult the EmbracePlus instruction for use for important information, warning, and cautions
	Indicates that EmbracePlus intentionally generates RF signals for data transmission to a connected device.

	Indicates the manufacturing date of the device
	Indicates that EmbracePlus is classified as IP67 - resistant to submersion in 1 m (about 3.3 feet) of water for up to 30 minutes
	Keep in dry conditions and away from rain
	Indicates the device Lot Number
	Indicates the minimum and maximum temperature at which the EmbracePlus shall be stored
	Indicates the minimum and maximum pressure at which the EmbracePlus shall be stored
	Indicates the minimum and maximum humidity at which the EmbracePlus shall be stored
	Indicates the need for the user to consult the EmbracePlus instruction for use for important information, warning, and cautions
	Indicates the web address at which the device instruction for use can be found

Intended use

EmbracePlus is intended to collect, process with embedded AI/ML algorithms, store, transfer and remote monitoring of the following* physiological parameters: Pulse Rate (PR), Pulse Rate Variability (PRV), Respiratory Rate (RR), Blood oxygen saturation (SpO2), Temperature (TEMP), Electrodermal Activity (EDA), Activity Counts, and Sleep Detection.

*Depending on sensor configuration

EmbracePlus is intended for temperature monitoring where monitoring temperature at the wrist is clinically indicated.

Indication for use statement

EMBRACEPLUS INDICATION FOR USE STATEMENT

EmbracePlus is intended for continuous patient monitoring of patients* in professional healthcare facilities, such as hospitals or skilled nursing facilities, or in their own home. EmbracePlus collected data are intended to be used for patient monitoring by trained healthcare professionals only.

*Patients' minimum age varies based on the country. The minimum is 2 years and up in the EU, and 18 years and up in the US.

EmbracePlus is intended for continuous monitoring of the following** physiological parameters in ambulatory patients who are 2 years of age or older :

- Pulse Rate;
- Pulse Rate Variability;
- Respiratory Rate;
- SpO2;
- Temperature;
- Electrodermal Activity (EDA);
- Activity Counts; and
- Sleep Detection.

*Depending on sensor configuration

EmbracePlus is intended for temperature monitoring where monitoring temperature at the wrist is clinically indicated.

Intended population and use environment

EmbracePlus is intended to be used by the general population (age 2 and up) that want/need to monitor their physiological parameters at home or in healthcare environments. Physiological Parameters might be reviewed by Caregivers, Healthcare providers, or Researchers.

Patients population and Medical Conditions

USER POPULATION

- Age: from 2 years of age (there might be limitation of wrist circumference for the wearable device) to elderly
- Weight: not relevant, but there might be limitation of wrist circumference
- Nationality: not relevant
- Patient status: the patient is often the same as the operator and should be willing and capable of managing its use

CAREGIVER POPULATION - EMBRACEPLUS

- Healthcare professionals
- Occupational doctors
- Nurses

Essential Performance Statement and Safety Information

EMBRACEPLUS ESSENTIAL PERFORMANCE

EmbracePlus shall be able to continuously collect, via specific sensors, the user's physiological parameters and transfer them to the paired companion app/system without being corrupted or lost. The maximum allowed quantity of data not being collected or lost during the normal use of the device is 5% of the duration of an acquisition session. An acquisition session is defined as a time period during which the user is correctly wearing a device with the purpose of acquiring data for reaching its intended use.

Evaluation of missing data are performed via a specific data portal by a healthcare professional.

The connection between EmbracePlus and the companion app/system can be lost for a maximum of 5% of the time, but the EmbracePlus should be able to automatically reconnect to the companion app/system.

The entire EmbracePlus is considered as an "Applied Part". EmbracePlus users shall be considered "Operators" in the meaning of IEC 60601-1.

EmbracePlus users can safely use all the EmbracePlus functions.

Service Life

EmbracePlus expected service life is 2 years.

Limitations

1. EmbracePlus is not intended as a primary diagnostic tool.
2. EmbracePlus is compatible only with iOS devices and Android devices that support Bluetooth 5.0.
3. Bluetooth connectivity will not work if EmbracePlus is submerged in water, and thus data cannot be transferred to the companion app/system.

Warnings and Precautions

WARNINGS

1. The user must wear the EmbracePlus correctly to ensure correct raw data acquisition and physiological parameter computation.
2. Do not clean EmbracePlus while wearing it.
3. Do not use EmbracePlus if it appears damaged. Check EmbracePlus for sharp edges and damage before each use.
4. Users who wear a pacemaker, or who have health conditions which might be sensitive to small electrical signals should always consult their physicians before wearing any electronic device, including EmbracePlus.
5. If EmbracePlus is used in a manner not specified by the manufacturer, their protection, functionalities, and performance they provide could be compromised.
6. Adult supervision is required. Due to the presence of small parts used in EmbracePlus, it is strongly advised that the device only be used on small children under adult supervision.
7. Use the EmbracePlus only with its original accessories provided by Empatica. Other cables and accessories might affect electromagnetic compatibility performances and device safety. Using EmbracePlus with accessories, transducers, or cables other than those specified, may also result in increased emissions or decreased immunity of Embrace.
8. Do not store EmbracePlus close to other electrical equipment.
9. Portable RF communications equipment, including antennas, can affect EmbracePlus, thus they can be used no closer than 30 cm (12 inches) to any part of EmbracePlus.

10. The EmbracePlus cable charger may generate heat when connected to a power source; be careful when handling it during the charge.
11. You must not wear EmbracePlus during either an MRI or an X-ray scan.
12. For the functioning of the system, the following conditions shall be met:
 - a. The user must wear the EmbracePlus properly and have it remain within close proximity (<30 feet or 10 meters) to the paired companion app/system;
 - b. The paired companion app/system must be connected to the internet;
 - c. The EmbracePlus does not show warning messages;
 - d. The EmbracePlus and the companion app/system must be charged and operating properly.
13. Only wear EmbracePlus on one of your wrists.

PRECAUTIONS

1. Do not place the EmbracePlus over broken or damaged skin.
2. Empatica does not recommend wearing the device if you have a known allergy to metals, or you are generally sensitive to skin contact with metals.
3. Keep EmbracePlus clean. Bacteria and dirt may cause skin itching or irritation.
4. Do not leave EmbracePlus in environments in which it may overheat beyond the recommended environmental limits (e.g., a car parked under the sun). If EmbracePlus is left exposed to high temperatures, allow it to cool before handling it to avoid possible skin burns.

5. Changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.
6. Avoid using EmbracePlus in a hot tub, sauna, or steam rooms.

Risks and Benefits

Risks

The list below represents all aspects that should be considered as cautions resulting from the Risk Management Process of the EmbracePlus. The user is required to carefully read this manual before using the device.

PHYSIOLOGICAL PARAMETERS PRESENCE AND QUALITY

Including any hazard that might result from sensors failure, electrical components damage, user errors or low algorithm performance. Low risk level – no significant harm – On-board memory, state of the art electrical design, fault detection algorithms and clear user experience have been implemented to mitigate the risk. Moreover, the EmbracePlus is not intended to diagnose a disease; only trained personnel may interpret the data.

PHYSIOLOGICAL PARAMETERS COMPUTATION

Including any hazard that might result from sensors failure or low algorithm performance. Low risk level – no significant harm – State of the art validation protocols have been performed to validate EmbracePlus physiological parameters. Moreover, the EmbracePlus is not intended to diagnose a disease; only trained personnel may interpret the data.

EMBRACEPLUS CLEANING

Including any hazard that might result from wrong cleaning procedure or use of non-compatible cleaning agents. Low risk level – no significant harm – State of the art validation protocols have been performed to validate EmbracePlus cleaning procedure and listed cleaning agents.

EMBRACEPLUS BIOCOMPATIBILITY

Including any hazard that might result from use of non-biocompatible construction material. Low risk level – no significant harm – EmbracePlus is made of a well-known biocompatible material, biocompatibility tests have been performed according to relevant regulatory standards.

EMBRACEPLUS SAFETY

Including any hazard that might result from electrical and mechanical failure. Low risk level – no significant harm – EmbracePlus has been designed, developed, and manufactured using state of the art techniques. In addition, safety tests according to relevant regulatory requirements have been performed.

EMBRACEPLUS ELECTROMAGNETIC COMPATIBILITY

Including any hazard that might result from internal or external interferences. Low risk level – no significant harm – EmbracePlus has been designed, developed, and manufactured using state of the art techniques. In addition, electromagnetic compatibility tests have been performed according to relevant regulatory requirements.

EMBRACEPLUS ENVIRONMENTAL CONDITIONS

Including any hazard that might result from the storage or use of EmbracePlus in extreme environmental conditions.. Low risk level – no significant harm – EmbracePlus has been designed, developed, and manufactured using state of the art techniques and materials that can withstand a wide range of environmental conditions. Indication of the environmental conditions in which the device shall be stored and used are presented in this manual.

EMBRACEPLUS USABILITY

Including any hazard that might result from the user errors or design failure. Low risk level – no significant harm – EmbracePlus user interface and user experience have been designed and tested following relevant regulatory requirements. Moreover, EmbracePlus labeling documents contain all the relevant information for a flawless use of the device.

EMBRACEPLUS CYBERSECURITY

Including any hazard that might result from data corruption and loss of data confidentiality. Low risk level – no significant harm – EmbracePlus software architecture includes state of the art controls to ensure data integrity.

EMBRACEPLUS FUNCTIONALITY

Including any hazard that might result from failures in device functionalities. Low risk level – no significant harm – EmbracePlus functionalities have been designed, developed, and tested using state of the art techniques.

Benefits

The known and potential benefits of EmbracePlus for clinical use are:

- The Remote monitoring capabilities enabled by EmbracePlus could reduce the caregivers’ risk of exposure to potential health risks.
- Facilitate users’ remote health status control without the need of hospitalization.

BENEFITS-RISK DETERMINATION

Based on the presented known benefits and the residual severity and probability of the identified risks, we determine EmbracePlus residual risks are outweighed by the potential benefits.

EmbracePlus Materials & Biocompatibility

EMBRACEPLUS COMPONENTS

EmbracePlus components relevant to the biocompatibility of the device are presented in the table below. This table presents the part name, the quantity per device, the manufacturing material, and the manufacturing process.

Part Name	Qty	Material	Production Method
POD Case	1	Polycarbonate	Injection molding
Thermal Ring	1	Stainless Steel 316L	Metal Injection Molding + CNC
PPG Window	1	Gorilla Glass 3	Molding and Ion-exchange
POD Buttons	2	Thermoplastic Polyurethane	Injection molding
E Ink Display Window	1	Gorilla Glass 3	Molding and Ion-exchange
Band	1	Biocompatible Silicone Rubber	Injection molding

Part Name	Qty	Material	Production Method
EDA electrodes	2	Stainless Steel 316L	CNC
Buckle	2	Stainless Steel 316L	CNC
Cable charger housing	1	Polycarbonate	Injection molding
Cable charger cap	1	Polycarbonate	Injection molding

BIOCOMPATIBILITY INFORMATION

Empatica takes great care in selecting the highest grade materials when designing its devices, so that users have the most secure and comfortable experience when using them. It is therefore unlikely that users will develop an allergic reaction from EmbracePlus. However, some people may notice a slight reaction on their skin. If this happens, please contact the entity that provided EmbracePlus to you with a picture of the affected area(s).

EmbracePlus Technical Specifications and Performance Characteristics

PERFORMANCE CHARACTERISTICS

Physiological Parameters measurement	Pulse Rate	Range: 24 - 240 bpm Resolution: 1 bpm Accuracy (no motion): 3 bpm _{Arms} Accuracy (motion): 5 bpm _{Arms}
	PRV - RMSSD	Typical range: 0 - 300 ms Resolution: 0.01 ms Accuracy: 98% of positive agreement, with respect to the ECG. Absolute relative error < 10%. The comparison was made on recordings of healthy subjects in still conditions.
	Respiration Rate	Range: 6 - 60 rpm Resolution: 1 rpm Accuracy: 3 rpm _{Arms}
	Temperature	Range: 0°C - +50°C Resolution: 0.1°C Accuracy: ± 0.1°C within 30.0°C - 45.0°C range
	Electrodermal Activity	Range: 0.01 µSiemens – 100 µSiemens Resolution: 1 digit ~ 900 pSiemens

Physiological Parameters measurement	Rest Detection	Range: 0 - 400 (0-99: wake epoch; 100-299: rest epoch; 300-399: rest interruption epoch; 400: for future use) Accuracy: The Rest detection algorithm did not miss any of the 46 PSG-derived sleep periods (Sensitivity = 100%). On average, the Rest detection algorithm detected an earlier sleep onset and a later sleep offset, with an overall longer sleep period duration compared to PSG.
	SpO2	Range: 70% - 100% Resolution: 1 % Accuracy: 2% Arms

TECHNICAL SPECIFICATIONS

EmbracePlus Case (POD)	Diameter: 32 mm / 1.26 in Thickness: 15 mm / 0.59 in Weight: 38 gr / 1.34 oz (including battery, band, and electronics) Color: Black with a matte finish Materials: Case in Polycarbonate and mechanical side buttons in Thermoplastic Polyurethane, PPG Glass (Gorilla Glass 3)
Ingress Protection Classification	EmbracePlus: IP 67 You can wear EmbracePlus while taking a shower, or in the rain. You can also wear it in pools (not in saltwater). You should not submerge EmbracePlus in water that is deeper than 3.3. feet or 1 meter or for more than 30 minutes. Avoid using EmbracePlus in a hot tub, sauna, or steam rooms. Please note that the Bluetooth connectivity will not work if EmbracePlus is submerged in water, and thus data cannot be transferred to the companion app/system. Recorded data will be transferred to the companion app/system as soon as the connection is re-established. EmbracePlus is dust tight, meaning dust will not enter the inner part of the device.
EmbracePlus Band	Min wrist circumference: 132 mm / 5.2 in Max wrist circumference: 195 mm / 7.68 in Width: 22.4 mm / 0.88 in Color: White upper face and teal back face Materials: Biocompatible silicone rubber and electrodes in stainless steel (SUS 316L) Features: Easily interchangeable band embedded with Empatica's custom made EDA sensors.

EmbracePlus Battery	Type: 3.8V Li-ion Capacity: 155 mAh Battery life: 2 years or up to 500 cycles Charging time (0% to 100%): 90 minutes (worst case) Battery duration (100% to 0%): 1 to 2 days (depends on the sensor configuration)
EmbracePlus Memory	Type: NOR Flash Memory Capacity: 128 MB Hours of data stored: 8 to 24 hours (depends on the sensor configuration)
E Ink Display	Type: Glass-protected E Ink display Display size: 25.9 mm Glass: Corning® Gorilla® Glass 3 Glass features: High resistance to scratch and sharp contact damage; high retained strength after use; superior surface quality E Ink specifications: Black & White, always-on, no backlight, 62 segments, 2Hz max refresh rate Display features: always-on clock face (time-synchronized with companion app/system), battery level (in percentage), Bluetooth® connection/disconnection indication, error states, menu navigation
Buttons	Type: two mechanical buttons Actions: menu navigation, confirmation, event tagging
Haptic Feedback	Type: Linear Resonant Actuator (LRA) vibration motor Features: Multi-pattern vibrations to signal success, errors, and give feedback to the user

Photoplethysmography (PPG) Sensor	Type: Custom-built by Empatica (multi-channel, multi-wavelengths) Channels: 4 acquisition channels (via 8 photodiodes) Wavelengths: Red, Infra-Red, Green via 9 LEDs Sampling Rate: 26 Hz – 208 Hz
Temperature Sensor	Type: Digital temperature sensor (Texas Instruments TMP117M) Range: 0 - 50°C Resolution: 0.01°C Accuracy: ± 0.1°C within 30.0°C - 45.0°C range (medical application calibration) Sampling Rate: 1 Hz - 4 Hz
Accelerometer and Gyroscope Sensor	Type: High precision 3D microelectromechanical accelerometer and gyroscope Range: ±16g, ±2000 dps Resolution: 16bit (0.488mg/LSB, 70mdps/LSB) Accuracy Accelerometer: ±0.01%/°C, 110µg/√Hz Accuracy Gyroscope: ±1% on sensitivity, ±0.007%/°C, 3.8mdps/√Hz Sampling Rate: 26 Hz – 208 Hz
Electrodermal Activity (EDA) Sensor	Type: Custom-built by Empatica (ventral side electrodes embedded in the band) Range: 0.01 µSiemens – 100 µSiemens Resolution: 1 digit ~ 55 pSiemens Sampling Rate: 1 - 4 Hz
Compatibility	iOS operative system: iOS 11 or higher Empatica supports iPhone 8 and higher
	Android 5.0 or higher Bluetooth® 5.0 is supported from Android 8.0 (Oreo) or higher (might depend on the specific manufacturer)

Connectivity	Main standard: Bluetooth® 5 Fallback standard: Bluetooth® Low Energy 4.2 Range: Maximum 10 meters / 30 feet (in line of sight) Radio Frequency: 2.4 - 2.5 GHz Maximum speed: Up to 2Mbits on supported Phones Security: AES 128 Bit (Advanced Encryption Standard)
--------------	--

STORAGE AND USE CONDITIONS

Storage Conditions	Temperature	-25°C to +70°C
	Humidity	20 – 95% Relative Humidity non condensing
	Pressure	1060 hPa to 500 hPa
	Battery preservation	EmbracePlus can be stored in its original box for a maximum of 3 months without charging. If EmbracePlus is not used for more than 90 days, make sure to charge it at least once every 3 months.
	After use storage	EmbracePlus should be properly stored. Do not expose EmbracePlus to direct sunlight, moisture, or rain.
Use Conditions	Temperature	(Use) -10°C to +40°C (Charging) 0°C to +40°C
	Humidity	20 – 95% Relative Humidity non condensing
	Pressure	1060 hPa to 700 hPa

NOTE: After storage at the extreme temperature indicated in the table above allow the device to return to room temperature (15°C – 30C°) before wearing EmbracePlus by leaving it at room temperature for at least 30 minutes.

DUST AND WATER RESISTANCE

EmbracePlus

You can wear EmbracePlus while taking a shower, or in the rain. You can also wear it in pools (not in saltwater). You should not submerge EmbracePlus in water that is deeper than 3.3. feet or 1 meter or for more than 30 minutes. Avoid using EmbracePlus in a hot tub, sauna, or steam rooms.

Please note that the Bluetooth connectivity will not work if EmbracePlus is submerged in water, and thus data cannot be transferred to the paired companion app/system. Recorded data will be transferred to the paired companion app/system as soon as the connection is re-established.

EmbracePlus is dust tight, meaning dust will not enter the inner part of the device.

COMPATIBILITY WITH MRI AND X-RAY SCANNING



IMPORTANT

You must not wear EmbracePlus during either an MRI or an X-ray scan.

During an X-ray, the doctor will usually ask you to remove all metal objects since they can affect the quality of the scan. For an MRI, it is especially important to remove all metal objects because the MRI’s magnets will attract them.

EMC Environment

GUIDANCE AND MANUFACTURER'S DECLARATION - ELECTROMAGNETIC EMISSIONS

EmbracePlus is intended for use in the electromagnetic environment specified below. The user of the EmbracePlus should assure that it is used in such an environment.

Emission Test	Compliance	Electromagnetic environment
RF emission CISPR 11	Group 1	EmbracePlus uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emission CISPR 11	Class B	The EmbracePlus device is suitable for use in domestic establishments and in establishments directly connected to the low voltage power supply network that supplies buildings used for domestic purposes.
Harmonic Emission EN 61000-3-2	Not Applicable	
Voltage Fluctuation/ flicker emission EN 61000-3-3	Not Applicable	

FCC COMPLIANCE

The EmbracePlus device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation (FCC Title 47, Subpart A, Part 15.19(3)).

Changes or modifications not expressly approved by the party responsible for compliance could void the user’s authority to operate the equipment (FCC Title 47, Subpart A, Part 15.21)

GUIDANCE AND MANUFACTURER'S DECLARATION - ELECTROMAGNETIC IMMUNITY

The EmbracePlus is intended for use in the electromagnetic environment specified below. The user of the EmbracePlus should assure that it is used in such an environment.

Immunity Test	IEC 60601 test level	Compliance Level	Electromagnetic environment- guidance
Electrostatic Discharge (ESD) EN 61000-4-2	8 kV contact 2/4/8/15 kV Air	IEC 60601-1-2 Test level	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30 %
Radiated electromagnetic field IEC 61000-4-3	10 V/m 80 MHz to 2.7 GHz	IEC 60601-1-2 Test level	Portable and mobile RF communications equipment should be used no closer to any part of EmbracePlus, including cables than the recommended minimum separation distance of 30 cm
Power Frequency (50/60 Hz) Magnetic Field EN 61000-4-8	30 A/m	30 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Interference may occur in the vicinity of equipment marked with the following symbol:



RECOMMENDED SEPARATION DISTANCE BETWEEN PORTABLE AND MOBILE RF COMMUNICATIONS EQUIPMENT AND EMBRACEPLUS

EmbracePlus is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The user of the EmbracePlus can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communication equipment (transmitter) and the EmbracePlus as recommended below, according to the maximum output power of the communication equipment.

Frequency Range and Level: RF wireless communication equipment

Test Frequency (MHz)	Modulation	Minimum IMMUNITY Level (V/m)	IMMUNITY Level Applied (V/m)
385	Pulse Modulation: 18 Hz	27	27
450	Pulse Modulation: 18 Hz	28	28
710 745 780	Pulse Modulation: 217 Hz	9	9
810 870 930	Pulse Modulation: 18 Hz	28	28
1720 1845 1970	Pulse Modulation: 217 Hz	28	28
2450	Pulse Modulation: 217 Hz	28	28
5240 5500 5785	Pulse Modulation: 217 Hz	9	9

MANUFACTURER INFORMATION

EmbracePlus is manufactured by

Name: Empatica S.r.L.

Address: Via Enrico Stendhal, 36

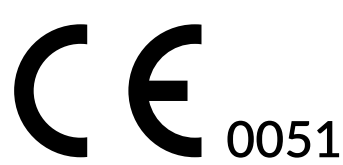
City: Milan

Zip: 20144

Phone: +39 02 36566473

Website: www.empatica.com

Made in Republic of Korea



ELECTRONIC LABELING INFORMATION

A PDF copy of this Instruction for Use is available at:

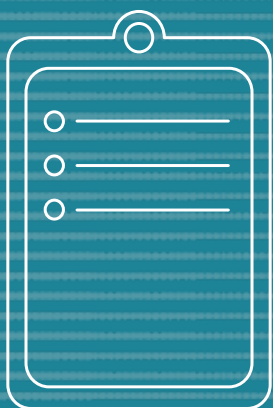
<http://empatica.com/manuals/embraceplus>

In case you want a paper copy of this Instruction for Use, you can request it from the administrator. The paper copy will be provided to you within 7 calendar days and free of charge.

.

07

Document history



Revision	Date	Description
Rev 1.0	April 1, 2021	Initial release of the document
Rev 2.0	April 16, 2021	Update
Rev 3.0	March 8, 2021	Update
Rev 4.0	November 03, 2021	Update